# Fine-Grained Crime Video Classification Using LV-MAE with Summary Token Refinement

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***Abstract*—Telling a slap from a punch or a push in surveillance video depends on how the motion unfolds, not on what a single frame shows. We study this fine-grained problem on our own dataset of 826 videos in 13 crime classes. Our model builds on LV-MAE, which learns long-video structure by masking and reconstructing a sequence of clip embeddings instead of pixels. Each video is split into 2-second segments that a frozen LanguageBind encoder turns into 1024-dimensional embeddings, and an LV-MAE encoder pretrained on these embeddings provides contextual segment features. On top of it we add a Summary Token module that refines each segment for the classification task, followed by temporal attention that weights the segments into one video-level feature. Pretraining takes about 21 minutes, compared with days for a pixel-level VideoMAE V2 baseline. On a test set of 168 videos, the proposed model reaches 80.95% accuracy at its best epoch, against 59.52% for the original LV-MAE with 5-second segments. A linear classifier on human-aware features reaches 74.24% accuracy and 0.697 macro F1 under a separate protocol.**

***Index Terms*—crime classification, surveillance video, masked autoencoder, LV-MAE, LanguageBind, temporal attention, fine-grained action recognition**

## I. Introduction

Surveillance cameras record far more footage than anyone can watch. Automatic analysis can help by flagging the clips that most likely contain a crime, but most existing work treats this as anomaly detection: it asks whether something unusual happens, not what exactly happened [10], [11]. For an operator, the difference matters. A slap, a punch and a push call for different responses, yet in any single frame they can look almost identical.

Fig. 1 shows the problem on a clip from our dataset. The decisive push lasts well under a second. Before it, two people simply face each other, and afterwards one of them is on the ground. A frame-level classifier cannot tell which violent act it is looking at, and a model that averages over the whole clip lets the quiet run-up and aftermath dilute the evidence. Recognising these classes therefore needs a model that keeps track of time and knows which moment to look at.

Two practical constraints shaped our approach. First, our dataset is small for a video transformer, with 826 clips in 13 classes. Self-supervised masked autoencoding [2]–[4] is the usual answer to limited labels, but our pixel-level VideoMAE V2 run was too slow to iterate on. Second, the classes depend on what people do, not on texture or lighting, which pixel reconstruction spends much of its capacity on. LV-MAE [1] addresses both: it masks and reconstructs a sequence of semantic clip embeddings, so pretraining is fast and the representation is already at the level of actions.

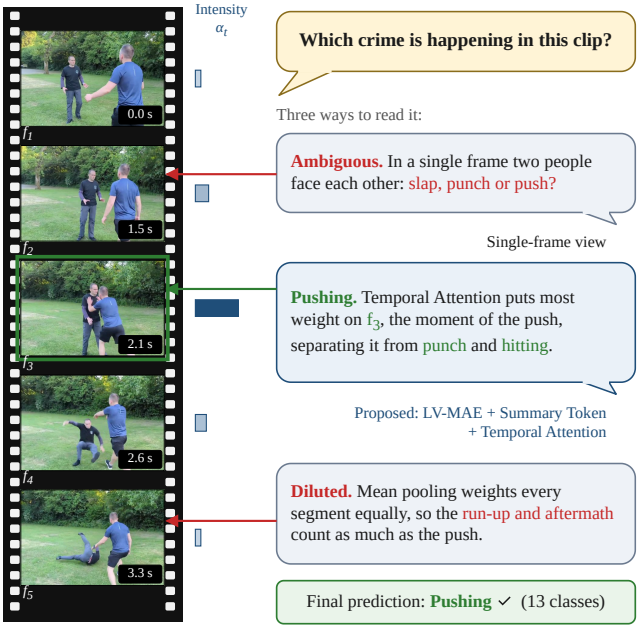



Fig. 1. Why temporal context matters. Five frames from a 3.5 s pushing clip. A single frame is ambiguous between slap, punch and push, and mean pooling lets the run-up and aftermath count as much as the push. The proposed model weights the segments (intensity $\alpha_t$, illustrative) and focuses on the push to predict *pushing*.

We adopt LV-MAE with LanguageBind [7] embeddings of 2-second segments as our base model and make the following contributions:

1) HA-LB Crime, a 13-class fine-grained crime video dataset of 826 clips, released on IEEE DataPort [19].
2) A Summary Token module, inserted after the pretrained

LV-MAE encoder, that refines each segment representation for classification, combined with temporal attention whose weights show which segments drive a prediction.
3) An experimental study covering segment length, pretraining cost against pixel-level VideoMAE V2, and a controlled ablation of the Summary Token.

## II. Related Work

### A. Masked Autoencoders for Video

MAE [4] introduced an asymmetric design: a heavy encoder sees only visible patches, a light decoder reconstructs the masked ones, and the decoder is discarded after pretraining. VideoMAE [2] extended this to video with tube masking at 90–95%. It showed that temporal redundancy makes very high masking ratios effective, even on small datasets. ST-MAE [5] reached the same conclusion with random spacetime masking. VideoMAE V2 [3] added dual masking, also masking the decoder input, which let it scale to a billion parameters. All of these reconstruct pixels over short clips of about 16 frames.

LV-MAE [1], our base model, moves masking from pixels to embeddings. It splits a long video into short clips and encodes each with a frozen multimodal encoder such as LanguageBind or InternVideo2 [8]. It then masks and reconstructs the embedding sequence. This decouples short-span appearance from long-span temporal structure, and it cuts pretraining cost substantially.

### B. Vision-Language Video Encoders

CLIP [6] aligned images and text through contrastive learning. LanguageBind [7] uses language as the binding modality for video, audio, depth and infrared, so its video embeddings are semantically organised. We use its video encoder, frozen, as the per-segment feature extractor.

### C. Surveillance Anomaly and Crime Recognition

Sultani et al. [10] introduced UCF-Crime and weakly supervised multiple-instance ranking for anomaly detection. RTFM [11] separated abnormal snippets by feature magnitude. More recent work brings in vision-language models: Vad-CLIP [12] adapts CLIP with a temporal adapter and learnable prompts, and TPWNG [13] uses text prompts to generate frame-level pseudo-labels. UCA [14] adds fine-grained event descriptions to UCF-Crime. These methods mostly output an anomaly score or align with coarse categories. Fine-grained classification among closely related violent acts has received less attention.

### D. Attention-Based Temporal Aggregation

Multi-head attention [9] underlies both of our modules. Set Transformer [15] pools a set with learnable seed queries, and Perceiver [16] lets a small latent array cross-attend to a large input. Video transformers [17], [18] aggregate with a learnable CLS token. Our head follows these ideas. Its contribution is placing learnable-query refinement and aggregation on top of LV-MAE encoder outputs for fine-grained crime classes.

TABLE I
Comparison with Related Video Datasets

| Dataset | Videos | Classes | Label type |
|---|---|---|---|
| Hockey Fight [23] | 1,000 | 2 | fight / no fight |
| Movies Fight [23] | 200 | 2 | fight / no fight |
| RWF-2000 [24] | 2,000 | 2 | violent / non-violent |
| UCF-Crime [10] | 1,900 | 13 | anomaly type, weak labels |
| XD-Violence [25] | 4,754 | 6 | violence type, weak labels |
| **HA-LB Crime (ours)** [19] | **826** | **13** | fine-grained interpersonal |

### E. Action Recognition and Violence Datasets

General action recognition moved from two-stream and 3D convolutional networks such as I3D [20] and SlowFast [21] to video transformers such as TimeSformer [18] and ViViT [22]. These models are trained on large datasets of everyday actions and need substantial compute and data to fine-tune. Violence-specific datasets are much smaller and usually binary. Hockey Fight and Movies Fight [23] ask only whether a clip contains a fight, and RWF-2000 [24] does the same for 2,000 real surveillance clips. UCF-Crime [10] and XD-Violence [25] cover more categories but label anomalies at the video level, and their classes mix interpersonal violence with events such as explosions or road accidents. Table I places our dataset, HA-LB Crime [19], among them. It is smaller than the anomaly benchmarks, but it is the only one in this list whose 13 classes are all fine-grained interpersonal actions, which is what makes classes like slap, punching and hitting hard to separate.

## III. Preliminaries: LV-MAE

LV-MAE [1] learns representations of long videos without labels (Fig. 2). A video is cut into $N$ short segments, and a frozen short-video encoder maps each segment to an embedding $\mathbf{z}_i$. The sequence $(\mathbf{z}_1, \ldots, \mathbf{z}_N)$ is then treated like a sentence of tokens. During pretraining, a random subset of the tokens is replaced by a learnable mask token. A Transformer encoder reads the visible tokens and produces $(\mathbf{h}_1, \ldots, \mathbf{h}_N)$, and a lighter Transformer decoder reconstructs the hidden embeddings $(\hat{\mathbf{z}}_1, \ldots, \hat{\mathbf{z}}_N)$. The training signal is the reconstruction error at the masked positions only.

This design differs from pixel-level video MAEs such as VideoMAE [2] in what it reconstructs. VideoMAE hides about 90% of the space-time patches of a 16-frame clip and predicts their pixels, which ties the model to a few seconds of video and to low-level appearance. LV-MAE predicts semantic embeddings instead, so a single token stands for a whole segment, the context can span minutes, and pretraining is much cheaper. After pretraining, the decoder is discarded and the encoder is reused for downstream tasks. Our method keeps this encoder and changes only what happens after it.

## IV. Proposed Approach

Our approach builds on the LV-MAE pipeline and adds a single trainable component, which we call the *Summary Token* module. It sits between the pretrained LV-MAE encoder and the temporal attention layer. Nothing else in the pipeline is

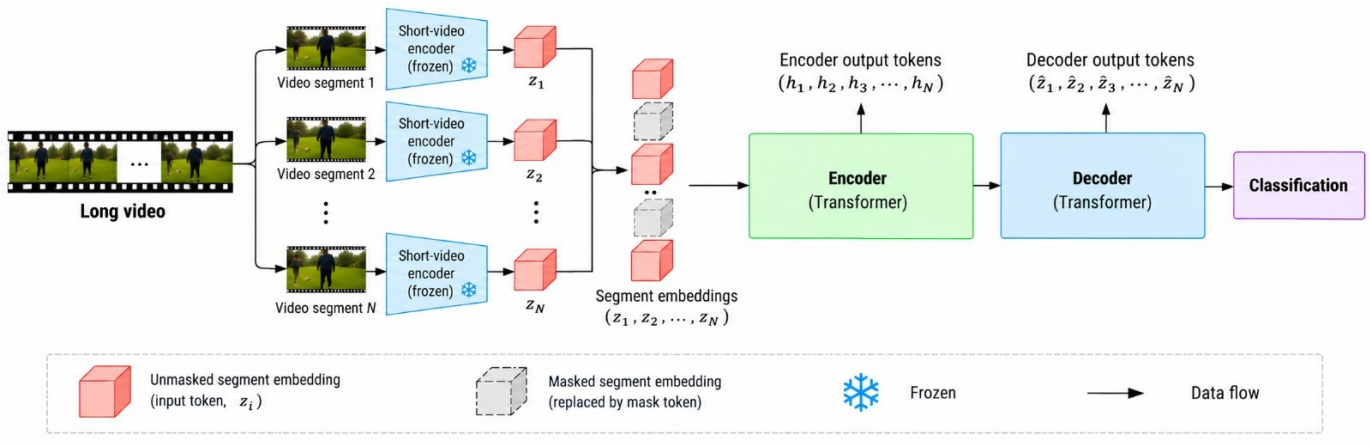



Fig. 2. LV-MAE [1], the base model. Video segments are encoded by a frozen short-video encoder into embeddings $\mathbf{z}_i$; masked embeddings are replaced by a mask token; a Transformer encoder and decoder reconstruct the masked embeddings. For classification the pretrained encoder outputs are used and the decoder is discarded.

replaced: LanguageBind still produces the segment features, LV-MAE still contextualises them, and the classifier is the same 13-way head.

### A. Segment Features and Pretraining

Fig. 3 shows the full system. Each video is split into consecutive 2-second segments $S_1, \ldots, S_n$, keeping the last partial segment. The frozen LanguageBind video encoder [7] maps every segment to a 1024-dimensional embedding. Clips of different length are padded to 20 segments, and a padding mask marks the empty positions.

We pretrain LV-MAE on these embeddings as described in Section III. The encoder has 12 Transformer layers, 8 heads and a width of 1024. The decoder has 8 layers, 16 heads and a width of 512. We mask 70% of the segments, so 6 of the 20 remain visible, and train with a mean-squared reconstruction loss on the masked positions only. We chose 70% rather than the 90% common in VideoMAE because, with only 20 tokens per clip, 90% would leave two visible segments, which is too little context to reconstruct anything meaningful. After pretraining, the encoder is applied to the full, unmasked sequence and returns one contextual vector $\mathbf{h}_i \in \mathbb{R}^{1024}$ per segment.

### B. Summary Token Module

The LV-MAE encoder was trained to reconstruct missing embeddings, not to recognise crimes. Its features are general, and some of what they encode, such as background, lighting or camera motion, is of little use to our classifier. The Summary Token module gives the network a small, trainable stage in which each segment can be reshaped for the task at hand before the segments are compared over time.

The module is shown in the centre of Fig. 3. It holds a learnable query vector $\mathbf{q}$, initialised randomly and updated by back-propagation like any other weight. For a segment $i$, the query attends to that segment's features through multi-head attention [9]. The result is added back to $\mathbf{h}_i$ and normalised. A small MLP then refines it further:

$$\tilde{\mathbf{s}}_i = \mathrm{LN}\big(\mathbf{h}_i + \mathrm{MHA}(\mathbf{q}, \mathbf{h}_i, \mathbf{h}_i)\big), \tag{1}$$

$$\mathbf{s}_i = \mathrm{LN}\big(\tilde{\mathbf{s}}_i + \mathrm{MLP}(\tilde{\mathbf{s}}_i)\big). \tag{2}$$

The same weights are shared by every segment, so a clip of $n$ segments still yields $n$ vectors, $\mathbf{s}_1, \ldots, \mathbf{s}_n$, each of dimension 1024. The module does not produce the video-level vector; that is left to the next stage. Unlike the LV-MAE decoder, which only exists during pretraining, the Summary Token is trained with the classification loss and is used at inference.

One point about the attention deserves care. In the current implementation the key and value for segment $i$ are the single vector $\mathbf{h}_i$, so the softmax has only one entry and its weight is always 1. The attention therefore does not select among frames; it acts as a learned projection of $\mathbf{h}_i$. This makes the module a refinement block rather than an attention mechanism over within-segment evidence. Feeding several tokens per segment would turn it into the latter, and we leave that to future work. The MLP inside the module expands the representation from 1024 to 2048 dimensions and projects it back to 1024, with a ReLU activation in between.

### C. Temporal Attention and Classification

The refined segments are combined by a temporal attention layer with a learnable CLS query $\mathbf{q}_c$ and 8 heads (right side of Fig. 3). Each segment receives a weight

$$\alpha_i = \mathrm{softmax}_i\left(\frac{\mathbf{q}_c^\top W_K \mathbf{s}_i}{\sqrt{d_k}}\right), \tag{3}$$

and the video feature $\mathbf{v}$ is the weighted sum of the projected segments. Padded positions are masked out before the softmax, so short clips are handled without special cases. Because the weights are stored for every clip, we can see which segments the model relied on. In the example of Fig. 1, we expect the weight to concentrate on the segment that contains the push.

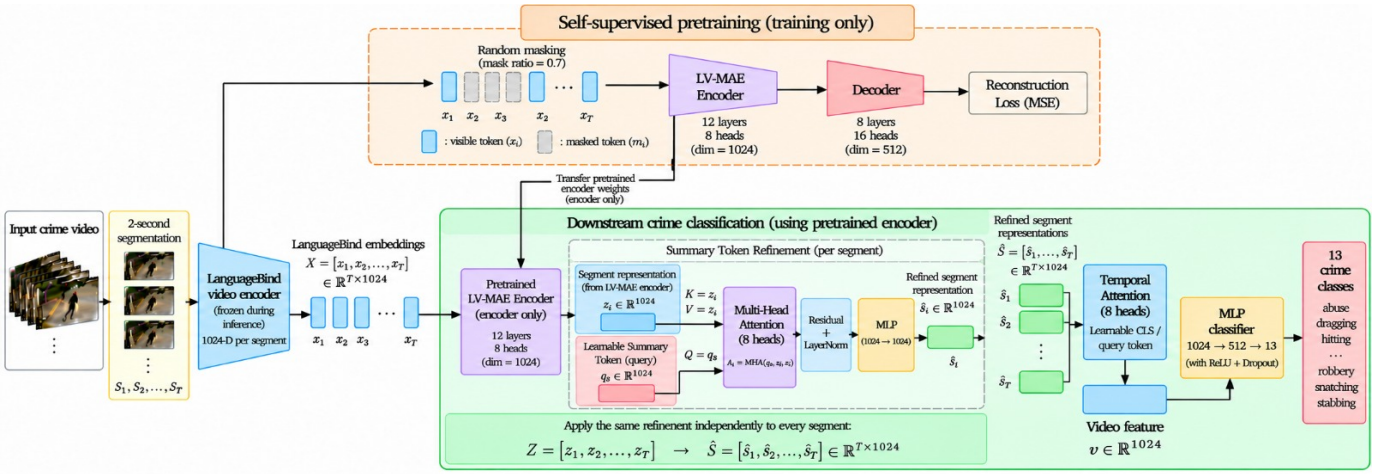



Fig. 3. Proposed architecture. Top: self-supervised LV-MAE pretraining (training only). LanguageBind segment embeddings are randomly masked (ratio 0.7), the encoder (12 layers, 8 heads, $D = 1024$) processes the visible tokens, and the decoder (8 layers, 16 heads, 512-D) reconstructs the masked embeddings under an MSE loss. Bottom: downstream crime classification. The pretrained encoder produces segment representations $\mathbf{z}_i$; the Summary Token module refines each segment with a learnable query $\mathbf{q}_s$, multi-head attention, a residual connection with LayerNorm, and an MLP; Temporal Attention with a learnable CLS query aggregates the refined segments into a video feature $\mathbf{v} \in \mathbb{R}^{1024}$, which the MLP classifier maps to 13 crime classes.

**Algorithm 1** Classifying a video

**Input:** video $V$; trained model
**Output:** predicted class

1: Split $V$ into 2-s segments and embed each with LanguageBind
2: Run the pretrained LV-MAE encoder to get $\mathbf{h}_1, \ldots, \mathbf{h}_n$
3: Refine each segment: $\mathbf{s}_i \leftarrow \mathrm{SummaryToken}(\mathbf{h}_i)$
4: Weight the segments with temporal attention to get $\mathbf{v}$
5: Classify $\mathbf{v}$ and return the highest-scoring of the 13 classes

TABLE II
CLASS DISTRIBUTION OF THE DATASET

| # | Class | Videos | # | Class | Videos |
|---|---|---|---|---|---|
| 1 | Abuse | 59 | 8 | Pushing | 71 |
| 2 | Dragging | 69 | 9 | Robbery | 58 |
| 3 | Hair Pulling | 54 | 10 | Shooting | 64 |
| 4 | Hitting | 69 | 11 | Slapping | 63 |
| 5 | Kicking | 61 | 12 | Snatching | 61 |
| 6 | Neck Grabbing | 60 | 13 | Stabbing | 74 |
| 7 | Punching | 63 | | **Total** | **826** |

The video feature goes through a two-layer classifier, 1024→512→13 with ReLU and a dropout of 0.5. The Summary Token, temporal attention and classifier are trained together with cross-entropy against the video label. LanguageBind stays frozen throughout, while the LV-MAE encoder is not frozen and is fine-tuned together with the new layers in Stage 2.

### D. Training and Inference

Training and inference follow the same path, summarised in Algorithm 1. We take care to keep the segment length, frame sampling, LanguageBind preprocessing and class-index mapping identical in both. The checkpoint stores the Summary Token weights together with the rest of the model, and we check that no keys are missing when it is loaded. Splits are made per video, so segments of one clip never appear in both training and test data.

## V. EXPERIMENTS

### A. Dataset

We collected our own dataset, HA-LB Crime [19], with 826 videos covering 13 fine-grained crime and violent-action classes (Table II). Each video carries one class label. The classes are roughly balanced, with 54 to 74 videos per class; hair pulling, robbery and abuse have slightly fewer than 60. Several classes are visually close, in particular slap, punching and hitting, which differ mainly in hand shape and motion, and robbery and snatching, which differ in the use of force. The videos were collected from movies, websites and YouTube Shorts and labelled into the 13 crime-activity categories following the dataset's annotation procedure. Of the 826 released videos, 823 were used in our experiments; the remaining three were excluded during preprocessing. We use a fixed split with 168 test videos and the remaining videos for training. The dataset is available on IEEE DataPort (doi: 10.21227/9ggn-cz58).

### B. Implementation Details

Table III lists the configuration of the proposed model. Embeddings are extracted once and cached, so pretraining and fine-tuning are separate runs over the same features, and an ablation only changes which head is attached to the encoder. We export the temporal attention weights of every test clip together with its true and predicted class for later inspection. The model is trained with the AdamW optimiser at a learning rate of $1 \times 10^{-5}$, using batch sizes of 32 and 64.

TABLE III
IMPLEMENTATION DETAILS

| Setting | Value |
|---|---|
| Segment length | 2 s, padded to 20 tokens |
| Segment encoder | LanguageBind video, frozen, 1024-D |
| LV-MAE encoder | 12 layers, 8 heads, width 1024 |
| Pretraining decoder | 8 layers, 16 heads, width 512 |
| Mask ratio | 70% (6 of 20 tokens visible) |
| Pretraining | 50 epochs, batch 16, weight decay 0.05 |
| Pretraining time | 21 min 35 s |
| Temporal attention | 8 heads, learnable CLS query |
| Classifier | 1024→512→13, ReLU, dropout 0.5 |
| Fine-tuning | 30 epochs, AdamW, lr $1 \times 10^{-5}$ |
| Encoder in Stage 2 | fine-tuned (not frozen) |
| Hardware | NVIDIA Tesla V100, 16 GB |

All experiments were run on an NVIDIA Tesla V100 GPU with 16 GB of memory.

*Pixel-level reference.* Before moving to LV-MAE we pretrained VideoMAE V2 [3] directly on video, with an importance-aware masking scheme of our own. Instead of masking tubelets at random, which can erase the one region that carries the event, we score every tubelet by combining RAFT optical flow [26] and U$^2$-Net saliency [27], $I_i = \alpha M_i + (1-\alpha)S_i$ with $\alpha = 0.7$, and mask exactly 90% of the tubelets by weighted selection, the same budget as standard VideoMAE. The run used 16 frames per clip, batch size 2 and mixed precision on a Tesla V100 with 16 GB. The masking idea worked, but each epoch also required running RAFT and U$^2$-Net over every clip, and the batch size was capped by memory, so pretraining was too slow to support the ablations we needed.

### C. Evaluation Protocol

To find out whether the Summary Token actually helps, we compare two models that differ only in this module (Fig. 6 and Table V). Model A passes the LV-MAE features straight to temporal attention, and model B inserts the Summary Token in between. Both use the same split, features, optimiser settings and training budget, and both are repeated with at least three seeds. Models are selected on the validation set, never on the test set. We report accuracy, macro precision, recall and F1, per-class recall and the confusion matrix; macro F1 is the main metric because the classes are not perfectly balanced.

### D. Results

*Pretraining cost.* Masking embeddings instead of pixels makes pretraining practical (Table VI). LV-MAE finished 50 epochs in 21 min 35 s, with a final reconstruction loss of $6.59 \times 10^{-5}$ on training and $6.31 \times 10^{-5}$ on validation data, so the pretext task does not overfit. The VideoMAE V2 run, by contrast, completed only 20 of 100 planned epochs in days of computation, and its loss was still falling. The two losses live in different spaces and are not directly comparable, but the difference in cost decided our direction.

*Main results.* Table VII summarises all runs. On our dataset, with the same 168 test videos, the original LV-MAE with 5-second segments reaches 59.52% accuracy and a macro F1 of 0.593. The proposed model, which uses 2-second segments, the Summary Token and temporal attention, reaches 80.95% at its best epoch and 76.79% after the last of 30 epochs (Fig. 5a). The gain of more than 17 points comes from two changes at once, the shorter segments and the new head. The planned A-versus-B ablation separates the two.

*Overfitting.* The proposed model fits the training set almost perfectly: at epoch 30 its training accuracy is 99.69% and its training loss 0.021, while the test loss stays at 1.406. The 4-point drop from the best to the final epoch confirms that the model keeps fitting the training data after test performance has peaked. This is expected with a few hundred training videos and a 12-layer encoder, and it is why our protocol selects models on a validation split rather than on the test set.

*Per-class behaviour.* The 5-second model is uneven across classes (Fig. 5b and Table IV). Shooting is recognised in all 13 test videos and hair pulling in 10 of 12, while hitting is recognised in only 3 of 14 and snatching in 5 of 13. The pattern fits our motivation. Shooting and hair pulling have distinctive poses or objects, whereas hitting overlaps visually with punching and slapping, and snatching with robbery. The human-aware classifier shows the same weaknesses: hitting has the second-lowest F1 (0.490), and punch the lowest (0.235), although punch has only 5 test samples in that split.

*Human-aware features.* A linear classifier trained on the human-aware features (Fig. 4) reaches a best validation macro F1 of 0.921 but a test accuracy of 74.24% and a test macro F1 of 0.697. The large gap between validation and test suggests that the validation set is easier than, or overlaps with, the training distribution, and this should be checked before the human-aware branch is combined with the proposed model. Because this run uses a different split of 458 test samples, its numbers are not directly comparable with the 168-video results.

*Segment length.* On an earlier version of the dataset, 2-second segments outperformed 5-second segments (84.85% vs. 80.56%), and on the current data the 5-second model is the weakest of all runs. A 5-second window can contain the run-up, the action and the aftermath together, which blurs exactly the detail that separates the fine-grained classes.

*What the attention weights show.* We export the temporal attention weight of every segment. In the clips we inspected, the weights are often close to uniform, for example about 0.37, 0.33 and 0.31 across the three segments of a 6-second clip. This is plausible for short clips in which the action spans most segments, but it also means the attention is not yet sharply localising the decisive moment, and that much of the gain may come from the per-segment refinement. Misclassified examples in the export include dragging predicted as neck grabbing, hair pulling and kicking, all involving close contact between two people.

## VI. DISCUSSION

### A. Why Embedding-Level Masking Suits This Task

Pixel-level masked autoencoders spend most of their capacity predicting texture, lighting and fine spatial detail. For

TABLE IV
PER-CLASS RESULTS OF THE LV-MAE 5-SECOND MODEL (168 TEST VIDEOS) AND THE HUMAN-AWARE LINEAR CLASSIFIER (458 TEST SAMPLES)

| | LV-MAE, 5-s segments | | Human-aware features, linear classifier | | | |
|---|---|---|---|---|---|---|
| Class | Correct / total | Accuracy (%) | Precision | Recall | F1 | Support |
| Abuse | 7 / 12 | 58.33 | 0.662 | 0.811 | 0.729 | 53 |
| Dragging | 9 / 14 | 64.29 | 0.500 | 0.588 | 0.541 | 34 |
| Hair pulling | 10 / 12 | 83.33 | 0.893 | 0.807 | 0.848 | 31 |
| Hitting | 3 / 14 | 21.43 | 0.632 | 0.400 | 0.490 | 30 |
| Kicking | 9 / 12 | 75.00 | 0.875 | 0.636 | 0.737 | 22 |
| Neck grabbing | 6 / 12 | 50.00 | 0.887 | 0.771 | 0.825 | 61 |
| Punch | 8 / 13 | 61.54 | 0.167 | 0.400 | 0.235 | 5 |
| Pushing | 6 / 13 | 46.15 | 0.868 | 0.868 | 0.868 | 38 |
| Robbery | 8 / 12 | 66.67 | 0.556 | 0.714 | 0.625 | 28 |
| Shooting | 13 / 13 | 100.00 | 0.959 | 0.922 | 0.940 | 51 |
| Slap | 9 / 13 | 69.23 | 0.708 | 0.850 | 0.773 | 20 |
| Snatching | 5 / 13 | 38.46 | 0.639 | 0.575 | 0.605 | 40 |
| Stabbing | 7 / 15 | 46.67 | 0.881 | 0.822 | 0.851 | 45 |
| Overall | 100 / 168 | 59.52 | accuracy 0.742 | | macro 0.697 | 458 |
| Macro / weighted average | | | 0.710 / 0.764 | 0.705 / 0.742 | 0.697 / 0.747 | |

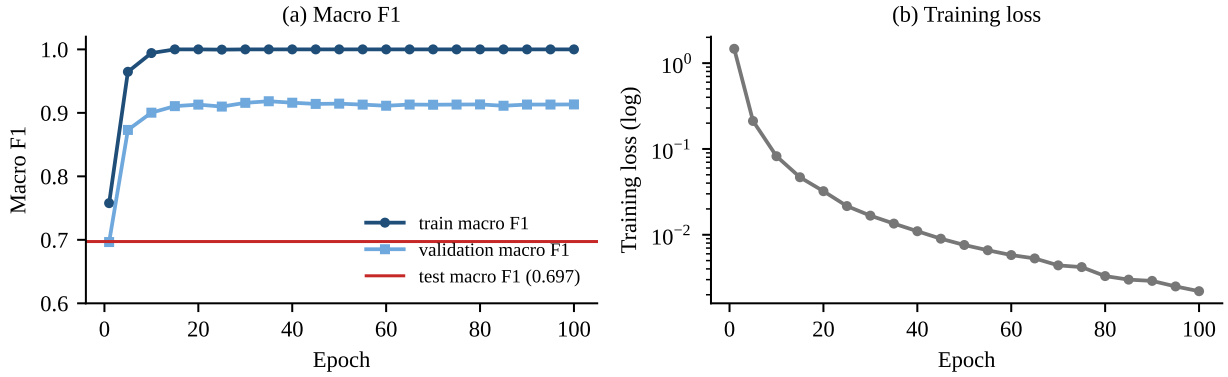



Fig. 4. Training of the human-aware linear classifier. (a) Training macro F1 reaches 1.0 by epoch 15 and validation macro F1 settles near 0.91, while test macro F1 is 0.697. (b) Training loss keeps falling over 100 epochs.

TABLE V
ABLATION CONFIGURATIONS

| Exp. | Configuration | Measures |
|---|---|---|
| A: Baseline | LV-MAE → Temporal Attention → classifier | reference accuracy, macro F1, per-class recall |
| B: Summary Token | LV-MAE → Summary Token → Temporal Attention → classifier | effect of the module (B vs. A) |
| C: Repeatability | A and B over ≥3 seeds | mean ± std of all metrics |

TABLE VI
PRETRAINING: PIXEL-LEVEL VS. EMBEDDING-LEVEL

| | VideoMAE V2 | LV-MAE (ours) |
|---|---|---|
| Masked unit | pixel tubelets | 2-s embeddings |
| Mask ratio | 90% | 70% of 20 tokens |
| Epochs completed | 20 of 100 | 50 of 50 |
| Wall-clock time | days (V100) | 21 min 35 s |

our classes, those details carry little of the label. Whether a clip shows a slap or a punch is decided by the shape of the hand and the arc of the movement over a fraction of a second, not by the colour of a wall. Reconstructing LanguageBind embeddings forces LV-MAE to model how semantic content changes from one segment to the next, which is closer to what the classifier needs. The practical benefit is just as important: with pretraining taking about twenty minutes, we could try segment lengths, masking ratios and heads that would each have cost days with VideoMAE V2.

### B. Choice of Mask Ratio

VideoMAE masks around 90% of its tokens because neighbouring video frames are highly redundant. Our tokens are 2-second segments, and a clip has at most 20 of them, so the redundancy between tokens is much lower. At 90%, only two segments would remain visible, and reconstructing the other eighteen would amount to guessing. We therefore use 70%, which leaves six visible segments. The final reconstruction

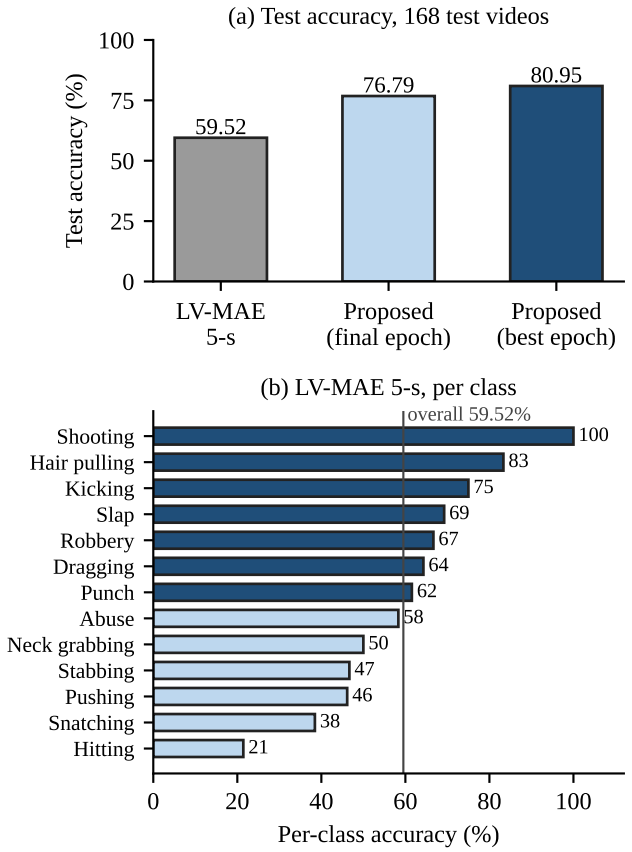



Fig. 5. (a) Test accuracy on the 168-video test set for the original LV-MAE with 5-second segments and for the proposed model at its final and best epochs. (b) Per-class accuracy of the 5-second LV-MAE; the vertical line marks its overall accuracy.

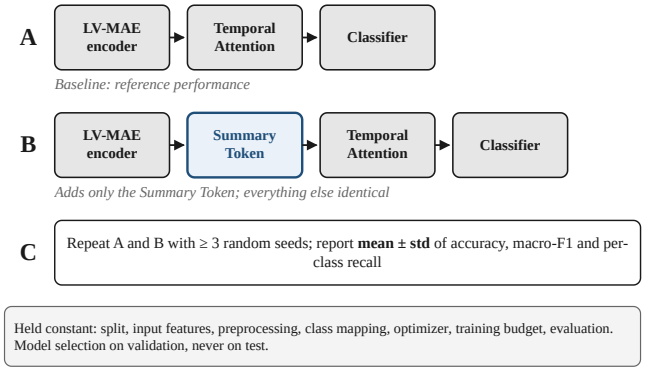



Fig. 6. Ablation design. A and B differ only in the Summary Token; C repeats both over several seeds.

losses, with validation slightly below training, suggest that the task is learnable without being trivial.

### C. Reading the Attention Weights

Table VIII lists exported attention weights for a few test clips. For most clips the weights are spread fairly evenly across

TABLE VII
CLASSIFICATION RESULTS

| Model | Test set | Acc. (%) | Macro F1 |
|---|---|---|---|
| *Our dataset* | | | |
| LV-MAE, 5-s segments | 168 videos | 59.52 | 0.593 |
| Proposed: LV-MAE 2-s + Summary Token + temporal attention | 168 videos | 80.95† | – |
| Human-aware features, linear classifier | 458 samples‡ | 74.24 | 0.697 |
| *Earlier version of the dataset* | | | |
| LV-MAE, 2-s segments | 132 videos | 84.85 | – |
| LV-MAE, 5-s segments | 144 videos | 80.56 | – |

†best test epoch; 76.79% at the final epoch. ‡separate split and protocol, not directly comparable.

TABLE VIII
EXPORTED TEMPORAL ATTENTION WEIGHTS FOR SAMPLE TEST CLIPS

| Clip | True / predicted | Segments (s) | Weights |
|---|---|---|---|
| 0 | Dragging / neck grabbing | 0–2, 2–4, 4–6 | 0.365, 0.326, 0.308 |
| 2 | Dragging / dragging | 0–2, 2–4, 4–6 | 0.351, 0.341, 0.308 |
| 5 | Dragging / kicking | 0–2, 2–4 | 0.523, 0.477 |
| 7 | Dragging / neck grabbing | 0–2, …, 6–8 | 0.255, 0.252, 0.255, 0.239 |

segments. A notable exception is a two-segment clip (video 5) in which the model puts 0.52 of the weight on the first segment and predicts kicking instead of dragging. Two lessons follow. First, near-uniform weights on short clips are expected, because the action fills most of the clip. Second, when the weights do become uneven, they show which segment drove a wrong decision, which makes errors easier to diagnose than with mean pooling.

### D. Error Analysis

The errors we see cluster around actions that involve close physical contact between two people. Dragging is confused with neck grabbing, hair pulling and kicking, all of which start with one person taking hold of another. Hitting is confused with punching and slapping, which share the same striking motion. Snatching overlaps with robbery, since both involve taking an object. These confusions suggest two directions. Person-centred features, such as those from our human-aware branch, may help separate who is acting on whom. Finer temporal resolution may help capture the brief moment that distinguishes one strike from another.

## VII. FUTURE WORK

Several extensions follow directly from our results. The most immediate is a validation-selected ablation over several seeds that separates the effect of shorter segments from that of the Summary Token. A second is to feed several tokens per segment, for example frame-level LanguageBind features, so that the Summary Token can attend within a segment rather than refining a single vector. A third is to fuse the human-aware features with the LV-MAE representation at the late-fusion point, which our error analysis suggests could help with

contact-heavy classes. Finally, we plan to evaluate the model on public violence benchmarks such as RWF-2000 and UCF-Crime to test how well the approach transfers beyond our own data.

## VIII. Limitations and Ethical Considerations

*Limitations.* Our dataset is small, with 54 to 74 videos per class, and a single test split of 168 videos means that one misclassified clip moves accuracy by about 0.6 points. The reported numbers come from single runs, and the best-epoch figure was selected on the test set, so they are optimistic; validation-based selection over several seeds is needed before the gains can be stated with confidence. The improvement of the proposed model over the 5-second LV-MAE combines two changes, shorter segments and the new head, and we have not yet separated them. Our runs also span more than one version of the data: an earlier version for the segment-length comparison, 823 videos for the main runs and 826 in the released dataset. Finally, as discussed in Section IV, when each segment is represented by a single vector the Summary Token acts as a learned refinement rather than true attention within a segment.

*Ethical considerations.* The videos show identifiable people in violent situations. Users of HA-LB Crime must follow its published license and the privacy, consent and redistribution conditions stated on the dataset page. A model of this kind can make mistakes that matter: in our results, hitting and snatching are recognised far less reliably than shooting. We therefore see it as a review-assist tool that ranks clips for a human operator, not as a system that makes decisions about people on its own, and the dataset must not be used to identify, track or profile individuals.

## IX. Conclusion

We presented a fine-grained crime video classifier built on LV-MAE, together with a new 13-class dataset of 826 clips. Masking semantic 2-second embeddings instead of pixels cut pretraining from days to about twenty minutes. Adding a Summary Token module and temporal attention on top of the pretrained encoder raised test accuracy from 59.52% for the original 5-second LV-MAE to 80.95% at the best epoch. Short segments consistently worked better than long ones, and the hardest classes were those that share most of their appearance, such as hitting, punching and slapping. Our next steps are a validation-selected ablation that separates the effect of segment length from that of the Summary Token, fusing the human-aware features with the proposed model, and feeding several tokens per segment so that the Summary Token can attend within segments.

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